

Solution Requirements

Date	06 July 2026
Team ID	
Project Name	Smart Lender — Applicant Credibility Prediction for Loan Approval
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Application Intake	System captures applicant details (gender, income, credit history, etc.) through a web form	High
2	Data Preprocessing	System cleans missing values, encodes categories, and scales numeric fields before prediction	High
3	Prediction Engine	System loads the trained XGBoost model and returns an Approved/Declined decision	High
4	Result Display	System renders a clear, immediate result to the user with the model's verdict	High
5	Model Training Pipeline	System supports retraining on updated datasets via the provided notebook	Medium
6	Model Comparison	System evaluates Decision Tree, Random Forest, KNN, and XGBoost and selects the best performer	Medium
7	Reporting	Notebook outputs accuracy, cross-validation scores, and classification reports per model	Medium
8	Other	Routes are exposed for home, application form, and result pages via Flask	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Prediction should return within 2 seconds of form submission	< 2 seconds per request
2	Scalability	App should handle moving from local Flask dev server to a hosted server	Supports concurrent local testing by all 5 team members
3	Security & Data Privacy	No applicant data is persisted	No PII stored on disk or DB

		beyond the single request/response cycle	
4	Reliability & Availability	App should run consistently across Windows/Anaconda environments	Verified working on 2+ team machines
5	Usability & Accessibility	Form should be simple, labeled clearly, and usable without ML knowledge	All 11 fields completable in under 2 minutes
6	Other	Model artifacts use a portable save format (.json) to avoid version mismatch errors	Loads successfully across XGBoost versions